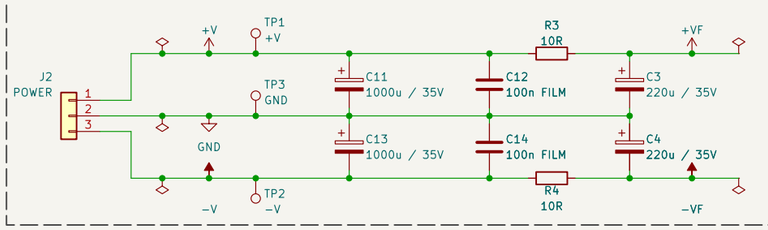
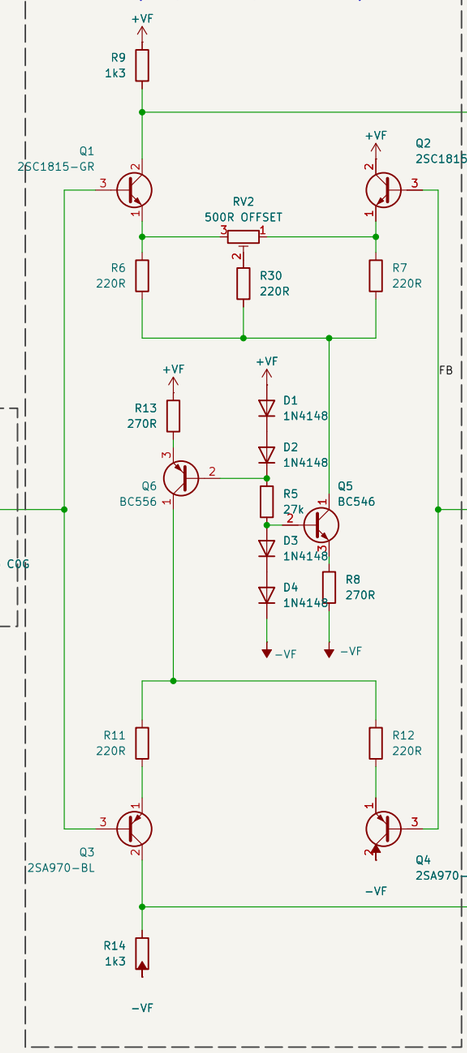
SUPPLY ENTRY / LOCAL DECOUPLING

C11–C14 at the output–pair emitters (Q12/Q13).

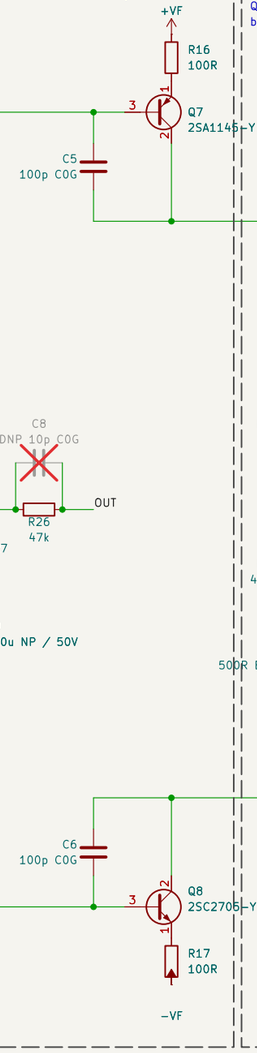
C3/C4 at the input stage and VAS.



INPUT STAGE (LTPs, TAIL CCS, OFFSET TRIM)

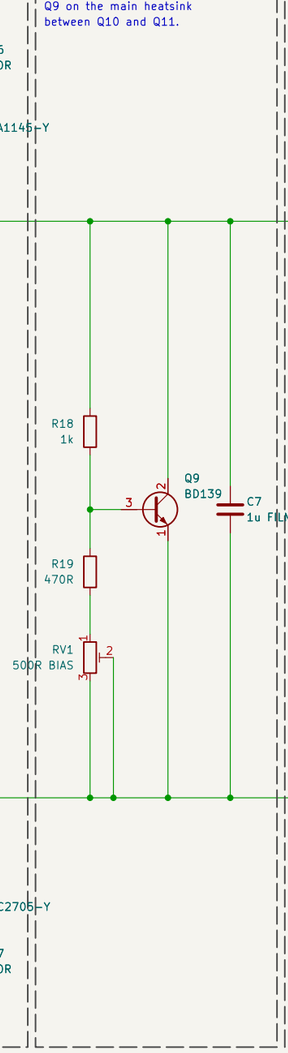


VAS / MILLER COMP. / FEEDBACK



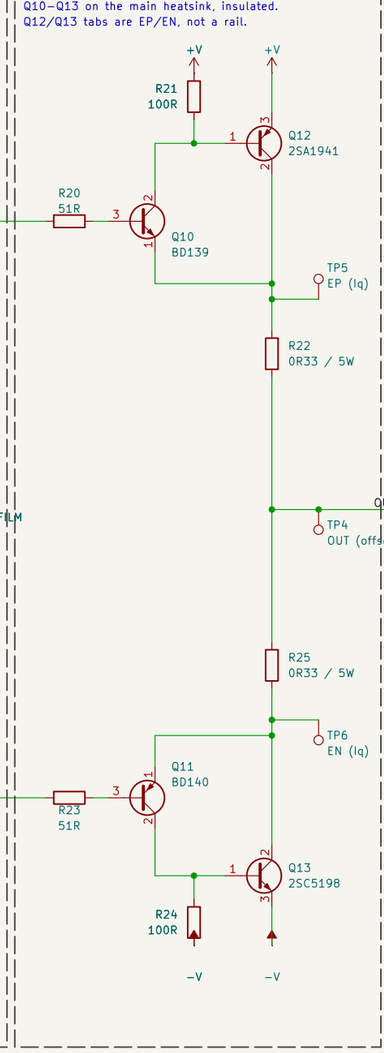
BIAS GENERATOR

Q9 on the main heatsink between Q10 and Q11.

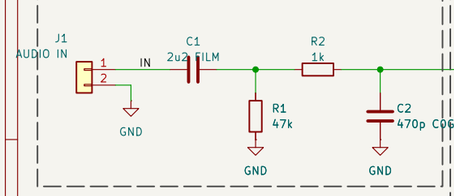


CFP OUTPUT STAGE

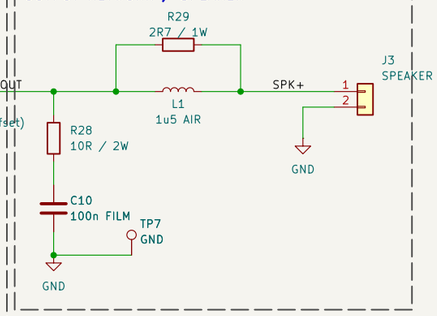
Q10–Q13 on the main heatsink, insulated. Q12/Q13 tabs are EP/EN, not a rail.



INPUT



OUTPUT NETWORK / SPEAKER



ClassA2 Rev A — Hiraga-inspired complementary Class-A amplifier, ONE MONO CHANNEL (build two boards).

- Topology: complementary LTP input (Q1/Q2, Q3/Q4) -> push-pull VAS (Q7/Q8, Miller C5/C6) -> VBE multiplier Q9 -> complementary-feedback-pair output (Q10+Q12, Q11+Q13).
- Rails: external +V/-V - +/-15.6 V loaded (12-0-12 VAC 100 VA, 216-253 V mains: -14.1 V valley to 20.3 V no-load). +VF/-VF = front-end rails via R3/C3, R4/C4.
- Bias: target 0.70 A per channel when warm = 231 mV across R22 (TP5-TP4) and R25 (TP4-TP6). Class A to ~7.5 W into 8 ohm; class AB above.
- Power-up: RV1 at maximum resistance (500R = minimum bias), current-limited supply, 8 ohm dummy load. Raise bias slowly. RV2 centred.
- Heatsink: Q9, Q10, Q11, Q12, Q13 on one insulated heatsink per channel, <=1.2 K/W (about 22 W per channel at idle). Q12/Q13 tabs are at EP/EN, NOT at a rail.
- Ground: one GND net. Return C9, R1, C2 and J1 screen to the input-ground point, C10/J3/C11-C14 to the power-ground point; join them at J2 pin 2.
- No output short-circuit protection. Speaker protection (relay, DC detect, turn-on delay >= 3 s) is external, between J3 and the speaker.
- Small C0G caps (C2, C5, C6, C8) use stocked 2012-metric SMD parts; through-hole C0G is a drop-in alternative at layout.

Generated by tools/build_schematic.py — rerun overwrites manual edits

Simulated with author-approximate BJT models; not bench verified

Schematic only — not released for PCB

Sheet: /

File: ClassA2.kicad_sch

Title: ClassA2 — Hiraga-inspired Class-A amplifier, mono channel

Size: A3	Date: 2026-10-26	**Rev: A (schematic review)**
KiCad E.D.A. 10.0.5		Id: 1/1